

(ii)

$$\rho_{150g} = \frac{m}{V}$$

$$V = \frac{0.36}{8.9 \times 10^3}$$

$$U = \rho \frac{0.36}{2(8.9 \times 10^3)} (9.81)$$

$$(mg - T)0.50 = (Mg - U)0.50$$

$$(0.250)(9.81) - (21)(0.015) = (0.36)(9.81) - \rho \frac{0.36}{2(8.9 \times 10^3)} (9.81)$$

$$\rho = 7.03 \times 10^3 \text{ kg m}^{-3}$$

(iii)

The metre rule is tilted anti-clockwise at an angle from its original position with less than half of block M submerged in the liquid.

OR

The metre rule is tilted at an angle from its original position with less than half of block M submerged in the liquid.

Additional information:

$$mg = Mg - U$$

$$(0.250)(9.81) = (0.36)(9.81) - b(7.0266 \times 10^3) \frac{0.36}{(8.9 \times 10^3)} (9.81)$$

$$b = 0.387$$

When the rule is balanced, 39% of block M is submerged in liquid.